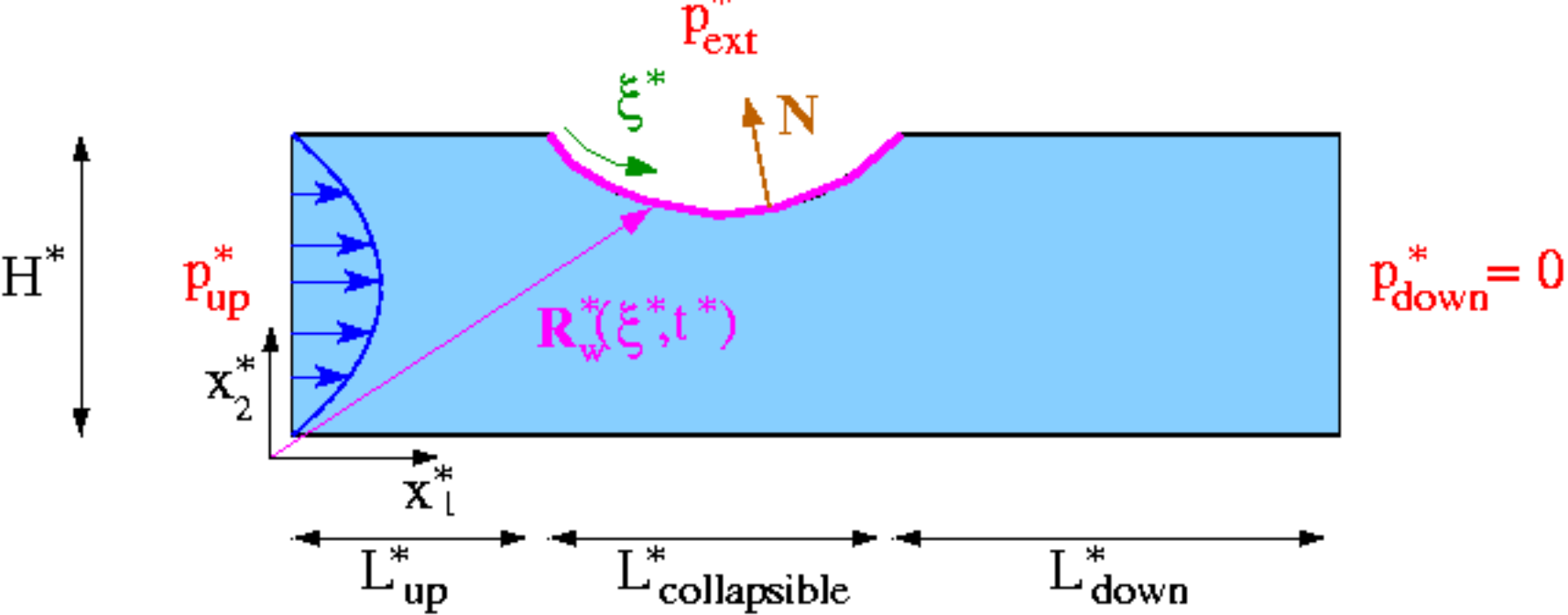
$p^*_{ext}$
$\xi^*$
N
$H^*$
$p^*_{up}$
$R^*_w(\xi^*,t^*)$
$p^*_{down}= 0$
$x^*_2$
$x^*_1$
$L^*_{up}$
$L^*_{collapsible}$
$L^*_{down}$